1 Rings and Ideals

IDEALS. PRIME AND MAXIMAL IDEALS

Proposition 1.1. *There is a one-to-one order-preserving correspondence between the ideals $\mathfrak{b}$ of A which contain $\mathfrak{a}$, and the ideals $\overline{\mathfrak{b}}$ of $A/\mathfrak{a}$, given by $\mathfrak{b} = \phi^{-1}\left(\overline{\mathfrak{b}}\right)$.*

Corollary 1.5. *Every non-unit of A is contained in a maximal ideal.*

Corollary 1.6. (i) *Let A be a ring and $\mathfrak{m} \neq (1)$ an ideal of A such that every $x \in A - \mathfrak{m}$ is a unit in A, Then A is a local ring and $\mathfrak{m}$ its maximal ideal.*

(ii) *Let A be a ring and $\mathfrak{m}$ a maximal ideal of A, such that every element of $1 + \mathfrak{m}$ is a unit in A, then A is a local ring.*

NILRADICAL AND JACOBSON RADICAL

We denote nilradical to be $\mathfrak{N}$ and Jacobson radical to be $\mathfrak{J}$.

Proposition 1.8. *The nilradical of A is the intersection of all the prime ideals of A.*

Sketch proof. Assume that $f \notin \mathfrak{N}$. Let Σ be a set of ideals $\mathfrak{a}$ with property $f^n \notin \mathfrak{a}$. Then Σ has maximal element which is prime. □

Proposition 1.9. *$x \in \mathfrak{J} \Leftrightarrow 1 - xy$ is a unit in A for all $y \in A$.*

OPERATIONS ON IDEALS

Two ideals $\mathfrak{a}$, $\mathfrak{b}$ are said to be coprime if $\mathfrak{a} + \mathfrak{b} = 1$. Thus we have $\mathfrak{a} \cap \mathfrak{b} = \mathfrak{a}\mathfrak{b}$.

Proposition 1.10 (Chinese Remainder Theorem)**.** *Define ϕ to be*

$$\phi : A \to \prod_{i=1}^{n} (A/\mathfrak{a}_i) , \qquad \phi(x) = (x + \mathfrak{a}_1, \cdots , \mathfrak{a}_n) .$$

(i) *If $\mathfrak{a}_i$ and $\mathfrak{a}_j$ are coprime whenever $i \neq j$, then $\mathfrak{a}_1 \cdots \mathfrak{a}_n = \bigcap_{i=1}^n \mathfrak{a}_i$.*

(ii) *ϕ is surjective $\Leftrightarrow$ $\mathfrak{a}_i$ and $\mathfrak{a}_j$ are coprime whenever $i \neq j$.*

(iii) *ϕ is injective $\Leftrightarrow$ $\bigcap_{i=1}^n \mathfrak{a}_i = (0)$.*

Proposition 1.11. *Denote $\mathfrak{a}$ to be ideal and $\mathfrak{p}$ to be prime ideal.*

(i) *If $\mathfrak{a} \subseteq \bigcup_{i=1}^n \mathfrak{p}_i$, then $\mathfrak{a} \subseteq \mathfrak{p}_i$ for some i.*

(ii) *If $\mathfrak{p} \supseteq \bigcap_{i=1}^n \mathfrak{a}_i$, then $\mathfrak{p} \supseteq \mathfrak{a}_i$ for some i. If $\mathfrak{p} = \bigcap_{i=1}^n \mathfrak{a}_i$, then $\mathfrak{p} = \mathfrak{a}_i$ for some i.*

The ideal quotient of $\mathfrak{a}$ and $\mathfrak{b}$ is

$$(\mathfrak{a} : \mathfrak{b}) = \{x \in A : x\mathfrak{b} \subseteq \mathfrak{a}\} .$$

The annihilator of $\mathfrak{b}$ is $\mathrm{Ann}(\mathfrak{b}) = (0 : \mathfrak{b})$.

The radical of $\mathfrak{a}$ is

$$\sqrt{\mathfrak{a}} = \{x \in A : x^n \in \mathfrak{a} \text{ for some } n > 0\} .$$

EXTENSION AND CONTRACTION

Let $f: A \to B$ be a ring homomorphism. Let $\mathfrak{a} \in A$ and $\mathfrak{b} \in B$. Define

$$\mathfrak{a}^e = Bf(\mathfrak{a}), \qquad \mathfrak{b}^c = f^{-1}(\mathfrak{b})$$

to be the extension of $\mathfrak{a}$ and the contraction of $\mathfrak{b}$, respectively. If $\mathfrak{b}$ is prime, $\mathfrak{b}^c$ is prime; if $\mathfrak{a}$ is prime, $\mathfrak{a}^e$ need not be prime.

Proposition 1.17. *$\mathfrak{a} \subseteq \mathfrak{a}^{ec}$, $\mathfrak{b} \supseteq \mathfrak{b}^{ce}$.*

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

Idempotent is an element e which satisfies $e^2 = e$.
A ring in which every element is an idempotent is a Boolean ring.

A topological space X is reducible if it can be written as a union $X = X_1 \cup X_2$ of two closed proper subsets X_1, X_2 of X. Equivalently, X is irreducible if every pair of non-empty open sets in X intersect, or if every non-empty open set is dense in X.

Let A be a ring and let $\operatorname{Spec}(A)$ be the set of all prime ideals of A. For each subset E of A, let $V(E)$ denote the set of all prime ideals of A which contain E. By Exercise 1.15, the sets $V(E)$ satisfy the axioms for closed sets in a topological space. The resulting topology on $\operatorname{Spec}(A)$ is called Zariski topology, and $\operatorname{Spec}(A)$ is called the prime spectrum of A.
For each $f \in A$, the complement of $V(f)$ in $\operatorname{Spec}(A)$, denote to be X_f, is called basic open set of $\operatorname{Spec}(A)$.
The subspace of $\operatorname{Spec}(A)$ consisting of the maximal ideals of A with the induced topology, is called the maximal spectrum of A and is denoted by $\operatorname{Max}(A)$.

Proposition (Exercise 1.27)**.** *Let k be algebraically closed. Then the affine space k^n with its classical Zariski topology is homeomorphic to* $\operatorname{Max}(k[x_1, \cdots, x_n])$.

2 Modules

Proposition 2.1. (i) *If $L \supseteq M \supseteq N$ are A-modules, then $(L/N)/(M/N) \cong L/M$.*

(ii) *If M_1, M_2 are submodules of M, then $(M_1 + M_2)/M_1 \cong M_2/(M_1 \cap M_2)$.*

An A-module is faithful if $\operatorname{Ann}(M) = 0$.

FINITELY GENERATED MODULES

Proposition 2.3. *M is a finitely generated A-module $\Leftrightarrow$ M is isomorphic to a quotient of A^n for some integer $n > 0$. More precisely, n is the number of generators of M and may not be unique.*

Proposition 2.4 (Cayley–Hamilton theorem)**.** *Let M be a finitely generated A-module, let $\mathfrak{a}$ be an ideal of A, and let ϕ be an A-module endomorphism of M such that $\phi(M) \subseteq \mathfrak{a}M$. Then ϕ satisfies an equation of the form*

$$\phi^n + a_1\phi^{n-1} + \cdots + a_n = 0$$

where the a_i are in $\mathfrak{a}$.

Proposition 2.6 (Nakayama's lemma)**.** *Let M be a finitely generated A-module and $\mathfrak{a}$ an ideal of A contained in the Jacobson radical of A. Then $\mathfrak{a}M = M$ implies $M = 0$.*

EXACT SEQUENCES

Proposition 2.10 (Snake lemma)**.** *Let*

$$\begin{array}{ccccccccc}
0 & \longrightarrow & M' & \xrightarrow{u} & M & \xrightarrow{v} & M'' & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f''} & & \\
0 & \longrightarrow & N' & \xrightarrow{u'} & N & \xrightarrow{v'} & N'' & \longrightarrow & 0
\end{array}$$

be a commutative diagram of A-modules and homomorphisms, with the rows exact. Then there exists an exact sequence

$$0 \longrightarrow \ker(f') \xrightarrow{\overline{u}} \ker(f) \xrightarrow{\overline{v}} \ker(f'') \xrightarrow{\delta} \operatorname{coker}(f') \xrightarrow{\overline{u}'} \operatorname{coker}(f) \xrightarrow{\overline{v}'} \operatorname{coker}(f'') \longrightarrow 0.$$

Let C be a class of A-modules and let λ be a function on C with values in $\mathbb{Z}$. The function λ is additive if for each short exact sequence

$$0 \to M' \to M \to M'' \to 0$$

in which all terms belong to C, we have $\lambda(M') - \lambda(M) + \lambda(M'') = 0$.

TENSOR PRODUCT

Proposition 2.12 (Universal property)**.** *Let M, N be A-modules. The tensor product of M and N over A is an A-module $M \otimes_A N$ together with an A-bilinear map*

$$f : M \times N \to M \otimes_A N, \qquad (m, n) \mapsto m \otimes n,$$

such that, for every A-module P and every A-bilinear map $\tau : M \times N \to P$, there exists a unique A-linear map $\widetilde{\tau} : M \otimes_A N \to P$ such that $\tau = \widetilde{\tau} \circ f$. Equivalently, the following diagram commutes:

$$\begin{tikzcd} M \times N \arrow[r, "f"] \arrow[dr, "\tau"'] & M \otimes_A N \arrow[d, dashed, "\widetilde{\tau}"] \\ & P \end{tikzcd}$$

Proposition 2.15. *Let A, B be rings, let M be an A-module, P a B-module and N an (A, B)-bimodule. then*

$$(M \otimes_A N) \otimes_B P \cong M \otimes_A (N \otimes_B P)\,.$$

Let M be an A-module and $f : A \to B$ be a homomorphism of rings. Define $M_B = B \otimes_A M$.

Proposition 2.19. *For an A-module N, the following are equivalent:*

(i) *The functor $T_N : M \mapsto M \otimes_A N$ on the category of A-modules is exact.*

(ii) *If $0 \to M' \to M \to M'' \to 0$ is any exact sequence of A-modules, then $0 \to M' \otimes N \to M \otimes N \to M'' \otimes N \to 0$ is exact.*

(iii) *If $f : M' \to M$ is injective, then $f \otimes 1 : M' \otimes N \to M \otimes N$ is injective.*

(iv) *If $f : M' \to M$ is injective and M, M' are finitely generated, then $f \otimes 1 : M' \otimes N \to M \otimes N$ is injective.*

An A-module N satisfying these properties is said to be a flat A-module.

ALGEBRA

Let $f : A \to B$ be a ring homomorphism. If $a \in A$ and $b \in B$, define a product

$$ab = f(a)b.$$

Thus, B has an A-module structure. The ring B is said to be an A-algebra.

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A partially ordered set I is said to be a directed set if for each pair $i, j \in I$ there exists $k \in I$ such that $i \leqslant k$ and $j \leqslant k$.

Let A be a ring, let I be a directed set and let $(M_i)_{i \in I}$ be a family of A-modules indexed by I. For each pair $i, j \in I$ such that $i \leqslant j$, let $\mu_{ij} : M_i \to M_j$ be an A-homomorphism, and suppose:

(1) μ_{ii} is the identity mapping of M_i, for all $i \in I$;

(2) $\mu_{ik} = \mu_{jk} \circ \mu_{ij}$ whenever $i \leqslant j \leqslant k$.

Then the modules M_i and homomorphisms μ_{ij} are said to form a direct system $\boldsymbol{M} = (M_i, \mu_{ij})$ over the directed set I.

Let D be the submodule of $\bigoplus M_i$ generated by all elements of the form $x_i - \mu_{ij}(x_i)$ whenever $i \leqslant j$ and $x_i \in M_i$. The direct limit of the direct system $\boldsymbol{M}$ is defined to be

$$\varinjlim M_i = \bigoplus_{i \in I} M_i \Big/ D\,.$$

Let $\mu : \bigoplus M_i \to \varinjlim M_i$ be the projection of quotient module. Define μ_i to be the restriction of μ to M_i.

A ring A (not necessarily commutative) is von Neumann regular if for every $a \in A$, there is an $x \in A$ with $a = axa$. A commutative ring A is absolutely flat if every A-module is flat. By Exercise 2.27, absolute flat and von Neumann regular are identical properties.

3 Rings and Modules of Fractions

MULTIPLICATIVELY CLOSED SETS

Proposition 3.3. *The operation S^{-1} is exact.*

Proposition 3.5. *Let M be an A-module. Then there exists a unique $S^{-1}A$ module isomorphism $f : S^{-1}A \otimes_A M \to S^{-1}M$.*

LOCAL PROPERTIES

Let P be a property of a ring A (or of an A-module M). Then the property P is said to be a local property, if A (or M) has $P \Leftrightarrow A_{\mathfrak{p}}$ (or $M_{\mathfrak{p}}$) has P for each prime ideal $\mathfrak{p}$ of A.

Proposition 3.8. *Let M, N be A-modules. Then the following are equivalent:*

(i) $M = 0$;

(ii) $M_{\mathfrak{p}} = 0$ *for all prime ideals* $\mathfrak{p}$ *of* A;

(iii) $M_{\mathfrak{m}} = 0$ *for all maximal ideals* $\mathfrak{m}$ *of* A;

Proposition 3.10. *Let M be A-modules. Then the following are equivalent:*

(i) M *is a flat* A*-module;*

(ii) $M_{\mathfrak{p}}$ *is a flat* $A_{\mathfrak{p}}$*-module for each prime ideal* $\mathfrak{p}$;

(iii) $M_{\mathfrak{m}}$ *is a flat* $A_{\mathfrak{m}}$*-module for each maximal ideal* $\mathfrak{m}$;

EXTENSIONS AND CONTRACTIONS OF FRACTIONS

If $\mathfrak{a}$ is an ideal in A, its extension $\mathfrak{a}^e$ in $S^{-1}A$ is $S^{-1}\mathfrak{a}$. Denote the contraction of $S^{-1}\mathfrak{a}$ in A to be $S(\mathfrak{a})$.

Proposition 3.11. (i) *Every ideal in $S^{-1}A$ is an extended ideal.*

(ii) *If $\mathfrak{a}$ is an ideal in A, then $S(\mathfrak{a}) = \mathfrak{a}^{ec} = \bigcup_{s \in S}(\mathfrak{a} : s)$.*

(iii) *$\mathfrak{a}$ is a contracted ideal $\Leftrightarrow$ no element of S is a zero-divisor in $A/\mathfrak{a}$.*

(iv) *Them prime ideal of $S^{-1}A$ are in one-to-one correspondence ($\mathfrak{p} \leftrightarrow S^{-1}\mathfrak{p}$) with the prime ideal of A which don't meet S.*

(v) *The operation S^{-1} commutes with formation of finite sums, products, intersections and radicals.*

Corollary 3.15. *If N, P are submodules of an A-module M and if P is finitely generated, then $S^{-1}(N : P) = \left(S^{-1}N : S^{-1}P\right)$.*

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A multiplicatively closed subset S of a ring A is said to be saturated if

$$xy \in S \Leftrightarrow x \in S \text{ and } y \in S.$$

Let M be an A-module. An element $x \in M$ is a torsion element of M if there exist a non-zero-divisor $a \in A$ such that $ax = 0$. All torsion elements of M form a submodule of M, which is called the torsion submodule of M and denote by $T(M)$. If $T(M) = 0$, the module M is said to be torsion free.

An A-module M is called faithfully flat if the tensor product functor $- \otimes_A M$ is exact and faithful. See Exercise 3.16.

Let M be an A-module. The support of M is defined to be the set $\mathrm{Supp}(M)$ of prime ideals $\mathfrak{p}$ such that $M_{\mathfrak{p}} \neq 0$.

$\mathrm{Spec}\,(A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \otimes_A B)$ is called the fiber of $f^* : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ over $\mathfrak{p}$.

4 Primary Decomposition

PRIMARY IDEALS

An ideal $\mathfrak{q} \in A$ is primary if $\mathfrak{q} \neq A$, and if $x \notin \mathfrak{q} \Rightarrow y^n \in \mathfrak{q}$ for some $n > 0$. This also means $x \notin \sqrt{\mathfrak{q}} \Rightarrow y \in \mathfrak{q}$.

Proposition 4.4. *Let $\mathfrak{q}$ be a $\mathfrak{p}$-primary ideal, $x \in A$ an element. Then*

(i) *if $x \in \mathfrak{q}$ then $(\mathfrak{q} : x) = (1)$;*

(ii) *if $x \notin \mathfrak{q}$ then $(\mathfrak{q} : x)$ is $\mathfrak{p}$-primary;*

(iii) *if $x \notin \mathfrak{p}$ then $(\mathfrak{q} : x) = \mathfrak{q}$;*

Proposition 4.8. *Let S be a multiplicatively closed subset of A, and let $\mathfrak{q}$ be a $\mathfrak{p}$-primary ideal.*

(i) *If $S \cap \mathfrak{p} \neq \varnothing$, then $S^{-1}\mathfrak{q} = S^{-1}A$.*

(ii) *If $S \cap \mathfrak{p} = \varnothing$, then $S^{-1}\mathfrak{q}$ is $S^{-1}\mathfrak{p}$-primary and its contraction in A is $\mathfrak{q}$.*

A primary decomposition of an ideal $\mathfrak{a}$ in A is an expression $\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i$.
If $\sqrt{\mathfrak{q}_i}$ are all distinct and $\mathfrak{q}_i \not\supseteq \bigcap_{j\neq i} \mathfrak{q}_j$, the primary decomposition is said to be irredundant.

UNIQUENESS

Proposition 4.5 (1st uniqueness theorem)**.** *Let $\mathfrak{a}$ be a decomposable ideal and let $\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i$ be a minimal primary decomposition of $\mathfrak{a}$. Let $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$. Then $\mathfrak{p}_i$ are precisely the prime ideals which occur in the set of ideals $\sqrt{(\mathfrak{a} : x)}$, and hence are independent of the particular decomposition of $\mathfrak{a}$.*

The prime ideals $\mathfrak{p}_i$ are said to belong to $\mathfrak{a}$, or to be associated with $\mathfrak{a}$. The minimal elements of the set $\{\mathfrak{p}_1, \cdots, \mathfrak{p}_n\}$ are called the minimal or isolated prime ideals belonging to $\mathfrak{a}$. The others are called embedded prime ideals.

Proposition 4.9. *Let S be a multiplicatively closed subset of A and let $\mathfrak{a}$ be a decomposable ideal. Let $\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i$ be a minimal primary decomposition of $\mathfrak{a}$. Let $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$ and suppose the $\mathfrak{q}_i$ numbered so that S meets $\mathfrak{p}_{m+1}, \cdots, \mathfrak{p}_n$ but not $\mathfrak{p}_1, \cdots \mathfrak{p}_m$. Then we have minimal primary decompositions*

$$S^{-1}\mathfrak{a} = \bigcap_{i=1}^{m} S^{-1}\mathfrak{q}_i, \qquad S(\mathfrak{a}) = \bigcap_{i=1}^{m} \mathfrak{q}_i.$$

A set Σ of prime ideals belonging to $\mathfrak{a}$ is said to be isolated if it satisfies the following condition: if $\mathfrak{p}'$ is a prime ideal beloning to $\mathfrak{a}$ and $\mathfrak{p}' \subseteq \mathfrak{p}$ for some $\mathfrak{p} \in \Sigma$, then $\mathfrak{p}' \in \Sigma$. Thus, every isolated set contains an isolated prime ideal.

Proposition 4.10 (2nd uniqueness theorem)**.** *Let $\mathfrak{a}$ be a decomposable ideal, let $\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i$ be a minimal primary decomposition of $\mathfrak{a}$, and let $\{\mathfrak{p}_{i_1}, \cdots, \mathfrak{p}_{i_m}\}$ be an isolated set of prime ideals of $\mathfrak{a}$. Then the ideal $\mathfrak{q}_{i_1} \cap \cdots \cap \mathfrak{q}_{i_m}$ is independent of the decomposition.*

Corollary 4.11. *The isolated primary components are uniquely determined by $\mathfrak{a}$.*

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

Let A be a ring and $\mathfrak{p}$ a prime ideal of A. The nth symbolic power of $\mathfrak{p}$ is defined to be the ideal $\mathfrak{p}^{(n)} = S_{\mathfrak{p}}(\mathfrak{p}^n)$.

5 Integral Dependence and Valuations

INTEGRAL DEPENDENCE

Proposition 5.1. *The following are equivalent:*

(i) *$x \in B$ is integral over A;*

(ii) *$A[x]$ is a finitely generated A-module;*

(iii) *$A[x]$ is contained in a subring C of B such that C is a finitely generated A-module;*

(iv) *There exist a faithful $A[x]$-module M which is finitely generated as an A-module.*

The set $\overline{A}$ of elements of B which are integral over A is called the integral closure of A in B. If $\overline{A} = A$, then A is said to be integrally closed in B. If $\overline{A} = B$, the ring B is said to be integral over A.

THE GOING-UP THEOREM

Corollary 5.9. *Let $A \subseteq B$ be rings, B integral over A. Let $\mathfrak{q}_1$, $\mathfrak{q}_2$ be prime ideals of B such that $\mathfrak{q}_1 \subseteq \mathfrak{q}_2$ and $\mathfrak{q}_1^c = \mathfrak{q}_2^c = \mathfrak{p}$. Then $\mathfrak{q}_1 = \mathfrak{q}_2$.*

Theorem 5.10 (Lying over property)**.** *Let $A \subseteq B$ be rings, B integral over A, and let $\mathfrak{p}$ be a prime ideal of A. Then there exists a prime ideal $\mathfrak{q}$ of B such that $\mathfrak{q} \cap A = \mathfrak{p}$.*

Theorem 5.11 (Going-up theorem)**.** *Let $A \subseteq B$ be rings, B integral over A. Let $\mathfrak{p}_1 \subseteq \cdots \subseteq \mathfrak{p}_n$ be a chain of prime ideals of A and $\mathfrak{q}_1 \subseteq \cdots \subseteq \mathfrak{q}_m$ $(m < n)$ a chain of prime ideals of B such that $\mathfrak{q}_i \cap A = \mathfrak{p}_i$ for $1 \leqslant i \leqslant m$. Then the chain $\mathfrak{q}_1 \subseteq \cdots \subseteq \mathfrak{q}_m$ can be extended to a chain $\mathfrak{q}_1 \subseteq \cdots \subseteq \mathfrak{q}_n$ such that $\mathfrak{q}_i \cap A = \mathfrak{p}_i$ for $1 \leqslant i \leqslant n$.*

THE GOING-DOWN THEOREM

Proposition 5.12. *Let $A \subseteq$ be rings, C the integral closure of A in B. Let S be a multiplicatively closed subset of A. Then $S^{-1}C$ is the integral closure of $S^{-1}A$ in $S^{-1}B$.*

An integral domain is said to be integrally closed (without qualification) if it is integrally closed in its field of fractions.

Proposition 5.13. *Let A be an integral domain. Then the following are equivalent:*

(i) *A is integrally closed;*

(ii) *$A_\mathfrak{p}$ is integrally closed, for each prime ideal $\mathfrak{p}$;*

(iii) *$A_\mathfrak{m}$ is integrally closed, for each maximal ideal $\mathfrak{m}$.*

Theorem 5.16 (Going-down theorem)**.** *Let $A \subseteq B$ be integral domains, A integrally closed, B integral over A. Let $\mathfrak{p}_1 \supseteq \cdots \supseteq \mathfrak{p}_n$ be a chain of prime ideals of A and $\mathfrak{q}_1 \supseteq \cdots \supseteq \mathfrak{q}_m$ $(m < n)$ a chain of prime ideals of B such that $\mathfrak{q}_i \cap A = \mathfrak{p}_i$ for $1 \leqslant i \leqslant m$. Then the chain $\mathfrak{q}_1 \supseteq \cdots \supseteq \mathfrak{q}_m$ can be extended to a chain $\mathfrak{q}_1 \supseteq \cdots \supseteq \mathfrak{q}_n$ such that $\mathfrak{q}_i \cap A = \mathfrak{p}_i$ for $1 \leqslant i \leqslant n$.*

VALUATION RINGS

Let B be an integral domain. B is a valuation ring of $\mathrm{Frac}(B)$ if, for each $x \neq 0$, either $x \in B$ or $x^{-1} \in B$ (or both).

Proposition 5.18. *Let B be a valuation ring, $K = \mathrm{Frac}(B)$ its field of fractions. Then*

(i) *B is a local ring;*

(ii) *If B' is a ring such that $B \subseteq B' \subseteq K$, then B' is a valuation ring of K;*

(iii) *B is integrally closed.*

Theorem 5.21. *Let K be a field, Ω an algebraically closed field. Let Σ be the set of all pairs (A, f), where A is a subring of K and $f : A \to \Omega$ a homomorphism. Define a partial order of Σ to be:*

$$(A, f) \leqslant (B, g) \Leftrightarrow A \subseteq B \text{ and } g|_A = f.$$

Then the maximal element of Σ is a valuation ring of K.

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A ring homomorphism $f : A \to B$ is said to have the going-up property if for $\mathfrak{p}_1 \subseteq \mathfrak{p}_2$ and $\mathfrak{p}_1 = f^{-1}(\mathfrak{q}_1)$, there exist a $\mathfrak{q}_2$ such that $\mathfrak{q}_1 \subseteq \mathfrak{q}_2$ and $f^{-1}(\mathfrak{q}_2) = \mathfrak{p}_2$. Reversing the inclusions of prime ideals we have the going-down property.

Proposition (Noether normalization lemma)**.** *Let k be a field and let $A \neq 0$ be a finitely generated k-algebra. Then there exist elements $y_1, \cdots, y_r \in A$ which are algebraically independent over k and such that A is integral over $k[y_1, \cdots, y_r]$.*

A ring A satisfied the following equivalent property is a Jacobson ring.

(i) Every prime ideal in A is an intersection of maximal ideals.

(ii) In every homomorphic image of A the nilradical is equal to the Jacobson radical.

(iii) Every prime ideal in A which is not maximal is equal to the intersection of the prime ideals which contain it strictly.

Let A be a valuation ring of a field K. The group U of units of A is a subgroup of the multiplicative group K^* of K. Let $\Gamma = K^*/U$. If $\xi, \iota \in \Gamma$ are represented by $x, y \in K$, define $\xi \geqslant \iota$ to mean $xy^{-1} \in A$. Then Γ is a totally ordered abelian group. It is called the value group of A.

Conversely, let Γ be a totally ordered abelian group. A valuation of K with values in Γ is a mapping $v : K^* \to \Gamma$ such that

(1) $v(xy) = v(x) + v(y)$,

(2) $v(x + y) \geqslant \min(v(x), v(y))$,

for all $x, y \in K^*$. Then the set of elements $x \in K^*$ such that $v(x) \geqslant 0$ is a valuation ring of K. This ring is called the valuation ring of v, and the subgroup $v(K^*)$ of Γ is the value group of v. Thus the concepts of valuation ring and valuation are equivalent.

6 Chain Conditions

Let Σ be a partially ordered set. Σ said to be satisfying the ascending chain condition (a.c.c) if, for every increasing sequence $x_1 \leqslant x_2 \leqslant \cdots$ in Σ there exists n such that $x_n = x_{n+1} = \cdots$.

Similarly, Σ said to be satisfying the descending chain condition (d.c.c) if, for every decreasing sequence $x_1 \geqslant x_2 \geqslant \cdots$ in Σ there exists n such that $x_n = x_{n+1} = \cdots$.

A module satisfying the ascending chain condition is said to be Noetherian. A module satisfying the descending chain condition is said to be Artinian.

Proposition 6.3. *Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of A-modules. Then*

(i) *M is Noetherian $\Leftrightarrow$ M' and M'' are Noetherian.*

(ii) *M is Artinian $\Leftrightarrow$ M' and M'' are Artinian.*

Proposition 6.5. *Let A be a Noetherian (resp. Artinian) ring, M a finitely generated A-module. Then M is Noetherian (resp. Artinian).*

A chain of submodules of a module M is a sequence (M_i) of submodules of M such that

$$M = M_0 \supsetneq M_1 \supsetneq \cdots \supsetneq M_n = 0.$$

The length of the chain is n. A composition series of M is a maximal chain, that is one which no extra submodules can be inserted, or equivalently M_{i-1}/M_i is simple.

Proposition 6.7. *Suppose that M has a composition series of length n. Then every composition series of M has length n, and every chain can be extended to a composition series.*

The length of a composition series of M is called the length of M, denoted as $l(M)$.

Proposition 6.10. *For a vector space, finite dimension $\Leftrightarrow$ finite length $\Leftrightarrow$ a.c.c. $\Leftrightarrow$ d.c.c.. If the conditions are satisfied, length equals dimension.*

DEFINITIONS AND PROPOSITIONS FROM EXERCISES

A topological space X is said to be Noetherian if the open subsets of X satisfy the ascending chain condition. Or equivalently, the closed subsets satisfy descending chain condition.

7 Noetherian Rings

Proposition 7.1. *Any homomorphic image of Noetherian ring is Noetherian.*

Proposition 7.2. *Let A be a subring of B; soppose that A is Noetherian and that B is finitely generated as an A-module. Then B is Noetherian.*

Theorem 7.5 (Hilbert's basis theorem)**.** *If A is Noetherian, then the polynomial ring $A[x]$ is Noetherian.*

Proposition 7.8. *Let $A \subseteq B \subseteq C$ be rings. Suppose that A is Noetherian, that C is finitely generated as an A-algebra and that C is either (i) finitely generated as a B-module or (ii) integral over B. Then B is finitely generated as an A-algebra.*

Proposition 7.9 (Zariski's lemma)**.** *Let k be a field, E a finitely generated k-algebra. If E is a field then it is a finite algebraic extension of k.*

PRIMARY DECOMPOSITION IN NOETHERIAN RINGS

Proposition 7.11. *In a Noetherian ring every ideal is a finite intersection of irreducible ideals.*

Proposition 7.12. *In a Noetherian ring every irreducible ideal is primary.*

Proposition 7.14. *In a Noetherian ring every ideal is contains a power of its radical.*

Proposition 7.15. *In a Noetherian ring the nilradical is nilpotent.*

Proposition 7.16. *Let A be a Noetherian ring, m a maximal ideal of A, $\mathfrak{q}$ any ideal of A. Then the following are equivalent:*

(i) $\mathfrak{q}$ *is* $\mathfrak{m}$*-primary;*

(ii) $\sqrt{\mathfrak{q}} = \mathfrak{m}$*;*

(iii) $\mathfrak{m}^n \subseteq \mathfrak{q} \subseteq \mathfrak{m}$ *for some* $n > 0$*.*

Proposition 7.17. *Let $\mathfrak{a} \neq (1)$ be an ideal in a Noetherian ring. Then the prime ideals which belong to $\mathfrak{a}$ are precisely the prime ideals which occor in the set of ideals $(\mathfrak{a} : x)$.*

8 Artinian Rings

Proposition 8.1. *In an Artinian ring every prime ideal is maximal.*

Proposition 8.3. *An Artinian ring has only a finite number of maximal ideals.*

The dimension of A is the supremum of the length of all chains of prime ideals in A.

Proposition 8.5. *A ring A is Artinian $\Leftrightarrow$ A is Noetherian and* $\dim(A) = 0$.

Proposition 8.6. *Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal. Then exactly one of the following statements is true:*

(i) $\mathfrak{m}^n \neq \mathfrak{m}^{n+1}$ *for all n;*

(ii) $\mathfrak{m}^n = 0$ *for some n. In this case A is an Artinian local ring.*

$\mathfrak{m}^n \neq \mathfrak{m}^{n+1}$ *for all n, or $\mathfrak{m}^n = 0$ for some n. In the latter case A is an Artinian local ring.*

Proposition 8.7 (Structure theorem of Artinian rings)**.** *An Artinaian ring A is uniquely (up to isomorphism) a finite direct product of Artinian local rings.*

Proposition 8.8. *Let A be an Artinian local ring. Then the following are equivalent:*

(i) *every ideal in A is principal;*

(ii) *the maximal ideal $\mathfrak{m}$ is principal;*

(iii) $\dim_{A/\mathfrak{m}} \left(\mathfrak{m}/\mathfrak{m}^2\right) \leqslant 1$.

9 Discrete Valuation Rings and Dedekind Domain

10 Completion

TOPOLOGIES AND COMPLETIONS

A topological abelian group G is both a topological space and an abelian group, with two structural mappings

$$\begin{aligned} G \times G \to G: &\quad (x, y) \mapsto x + y \\ G \to G: &\quad x \mapsto -x \end{aligned}$$

being continuous.

If $\{0\}$ is closed in G, then G is Hausdorff.

Since $T_a(x) = x + a$ is a homeomorphism, the topology of G is uniquely determined by the neighborhoods of 0 in G.

Proposition 10.1. *Let H be the intersection of all neighborhoods of 0 in G. Then*

(i) *H is a subgroup.*

(ii) *H is the closure of $\{0\}$.*

(iii) *G/H is Hausdorff.*

(iv) *G is Hausdorff $\Leftrightarrow$ $H = 0$.*

Assume that $0 \in G$ has a countable fundamental system of neighborhoods. A Cauchy sequence (x_ν) of G is that, for any neighborhood U of 0, there exists an integer $s(U)$ such that $x_m - x_n \in U$ for all $m, n \geqslant s(U)$.

Let $C(G)$ be the set of all Cauchy sequences in G. Define an equivalence relation $\sim$ on $C(G)$ where $(x_\nu) \sim (y_\nu)$ means $x_n - y_n \to 0$ in G. The completion $\hat{G}$ of G is the set of all equivalence classes in $C(G)/\sim$.

For each $x \in G$, define a homomorphism $\phi : G \to \hat{G}$ maps to a constant sequence (x). Then ϕ is injective iff G is Hausdorff.

Assume further that $0 \in G$ has a fundamental system of neighborhoods consisting of a sequence of subgroups

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n \supseteq \cdots .$$

For any $(x_\nu) \in C(G)$ and a fixed n, the image sequence $(\xi_\nu + G_n)$ in G/G_n is ultimately constant. Let ξ_n be this ultimate constant value.

As n varies, the projection $\theta_{n+1} : G/G_{n+1} \to G/G_n$ satisfies $\theta_{n+1}(\xi_{n+1}) = \xi_n$ for all n. The groups G/G_n combined with maps θ_n form an **inverse system**, and the group of all such coherent sequences (ξ_ν) is called the **inverse limit**, denoted as

$$\hat{G} = \varprojlim G/G_n.$$

A direct (or inverse) system is called injective/surjective system, if every transition map μ_{ij} is injective/surjective.

Proposition 10.2. *If $0 \to \{A_n\} \to \{B_n\} \to \{C_n\} \to 0$ is an exact sequence of inverse system, then*

$$0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n$$

is exact. If $\{A_n\}$ is surjective, then

$$0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n \to 0$$

is exact.

Corollary 10.3. *Let $0 \to G' \to G \to G'' \to 0$ be an exact sequence of groups. Let G have the topology defined by a sequence $\{G_n\}$ of subgroups, and give G' and G'' the topologies by the contraction and extension of $\{G_n\}$, respectively. Then*

$$0 \to \hat{G}' \to \hat{G} \to \hat{G}'' \to 0$$

is exact.

Corollary 10.4. *$\hat{G}_n$ is a subgroup of $\hat{G}$ and $\hat{G}/\hat{G}_n \cong G/G_n$.*

Corollary 10.5. *$\hat{\hat{G}}_n \cong \hat{G}_n$.*

If $G \to \hat{G}$ is an isomorphism, G is said to be complete.

Let A be a ring, $\mathfrak{a}$ its ideal. The topology defined by $G_n = \mathfrak{a}^n$ ($G_0 = A$) is called the $\mathfrak{a}$-adic topology, and the ring A combined with this topology is a topological ring.

FILTRATIONS

A chain $M = M_0 \supseteq M_1 \supseteq \cdots \supseteq M_n \cdots$ is called a filtration of M, and denoted by (M_n). It is an $\mathfrak{a}$-filtration if $\mathfrak{a}M_n \subseteq M_{n+1}$ for all n, and a stable $\mathfrak{a}$-filtration if $\mathfrak{a}M_n = M_{n+1}$ for all sufficiently large n.

The $\mathfrak{a}$-adic completion of M is defined to be

$$\hat{M} = \varprojlim M/\mathfrak{a}^n M.$$

GRADED RINGS AND MODULES

A graded ring is a ring A together with a family $(A_n)_{n\geqslant 0}$ of additive subgroups of A, such that $A = \bigoplus_{n=0}^{\infty} A_n$ and $A_m A_n \subseteq A_{m+n}$ for all $m, n \geqslant 0$.

Proposition 10.7. *A graded ring A is Noetherian, iff A_0 is Noetherian and A is finitely generated as an A_0-algebra.*

Proposition 10.9 (Artin-Rees lemma)**.** *Let A be a Noetherian ring, $\mathfrak{a}$ and ideal in A, M a finitely-generated A-module, (M_n) a stable $\mathfrak{a}$-filtration of M. If M' is a submodule of M, then $(M' \cap M_n)$ is a stable $\mathfrak{a}$-filtration of M'.*

Proposition 10.12. *Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of finitely-generated modules over a Noetherian ring A, then the sequence of $\mathfrak{a}$-adic completion $0 \to \hat{M}' \to \hat{M} \to \hat{M}'' \to 0$ is exact.*

Proposition 10.13. *If M is finitely-generated A-module, $\hat{A} \otimes_A M \to \hat{M}$ is surjective. If A is Noetherian, then $\hat{A} \otimes_A M \to \hat{M}$ is an isomorphism.*

Proposition 10.14. *If A is a Noetherian ring, the $\mathfrak{a}$-adic completion $\hat{A}$ is a flat A-algebra.*

Proposition 10.15. *If A is a Noetherian ring, $\hat{A}$ its $\mathfrak{a}$-adic completion, then*

(i) *$\hat{\mathfrak{a}} = \hat{A}\mathfrak{a} \cong \hat{A} \otimes_A \mathfrak{a}$;*

(ii) *$\widehat{(\mathfrak{a}^n)} = (\hat{\mathfrak{a}})^n$;*

(iii) *$\hat{\mathfrak{a}}^n/\hat{\mathfrak{a}}^{n+1} \cong \mathfrak{a}^n/\mathfrak{a}^{n+1}$;*

(iv) *$\hat{\mathfrak{a}}$ is contained in the Jacobson radical of $\hat{A}$.*

Proposition 10.16. *Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal. Then the $\mathfrak{m}$-adic completion $\hat{A}$ of A is a local ring with maximal ideal $\hat{\mathfrak{m}}$.*

Proposition 10.17 (Kull's theorem)**.** *Let A be a Noetherian ring, $\mathfrak{a}$ an ideal, M a finitely-generated A-module and $\hat{M}$ the $\mathfrak{a}$-adic completion of M. Then the kernel $E = \bigcap_{n=1}^{\infty} \mathfrak{a}^n M$ of $M \to \hat{M}$ consists of those $x \in M$ annihilated by some element of $1 + \mathfrak{a}$.*

THE ASSOCIATED GRADED RING

Let A be a ring and $\mathfrak{a}$ an ideal of A. The associated graded ring is the graded ring

$$G(A)\,(=G_{\mathfrak{a}}(A)) = \bigoplus_{n=0}^{\infty} \mathfrak{a}^n/\mathfrak{a}^{n+1} \qquad (a^0 = A).$$

Similarly, if M is an A-module and (M_n) is an $\mathfrak{a}$-filtration of M, then the associated graded module is

$$G(M) = \bigoplus_{n=0}^{\infty} M_n/M_{n+1},$$

which is a graded $G(A)$-module. Let $G_n(M) = M_n/M_{n+1}$.

Proposition 10.22. *Let A be a Noetherian ring, $\mathfrak{a}$ an ideal of A. Then*

(i) *$G_{\mathfrak{a}}(A)$ is Noetherian;*

(ii) *$G_{\mathfrak{a}}(A)$ and $G_{\hat{\mathfrak{a}}}(\hat{A})$ are isomorphic as graded rings;*

(iii) *If M is a finitely-generated A-module and (M_n) is a stable $\mathfrak{a}$-filtration of M, then $G(M)$ is a finitely-generated graded $G_{\mathfrak{a}}(A)$-module.*

Proposition 10.23. *Let $\phi : A \to B$ be a homomorphism of filtered groups, i.e. $\phi(A_n) \subseteq B_n$, and let $G(\phi) : G(A) \to G(B)$, $\hat{\phi} : \hat{A} \to \hat{B}$ be the induced homomorphisms of the associated graded and completed groups. Then*

(i) *$G(\phi)$ injective $\Rightarrow$ $\hat{\phi}$ injective;*

(ii) *$G(\phi)$ surjective $\Rightarrow$ $\hat{\phi}$ surjective.*

Proposition 10.26. *If A is a Noetherian ring, $\mathfrak{a}$ an ideal of A, then the $\mathfrak{a}$-adic completion $\hat{A}$ of A is Noetherian.*

11 Dimension Theory

HILBERT FUNCTIONS

Let A be a Noetherian graded ring and M be a finitely-generated graded A-module. Let λ be an additive function on the class of all finitely generated A_0-modules. The Poincaré series of M is the power series

$$P(M,t) = \sum_{n=0}^{\infty} \lambda(M_n)\, t^n \qquad \in \mathbb{Z}[[t]].$$

Theorem 11.1 (Hilbert-Serre theorem)**.** *$P(M,t)$ is a rational function in t of the form $f(t)/\prod_{i=1}^{d}\left(1-t^{k_i}\right)$, where $f(t) \in \mathbb{Z}[t]$.*

The order of the pole of $P(M,t)$ at $t=1$ is denoted as $d(M)$.

Corollary 11.2. *If each $k_i = 1$, then for all sufficient large n, $\lambda(M_n)$ is a polynomial in n. More specifically, if $f(t) = \sum_{k=0}^{N} a_k t^k$, we have*

$$\lambda(M_n) = \sum_{k=0}^{n} a_k \binom{d+n-k-1}{d-1}.$$

This is called Hilbert function or Hilbert polynomial of M, and its leading term is $\left(\sum a_k\right) n^{d-1}/(d-1)!$.

Proposition 11.3. *If $x \in A_k$ is not a zero-divisor in M then $d(M/xM) = d(M) - 1$.*

Proposition 11.4. *Let A be a Noetherian local ring, $\mathfrak{m}$ its maximal ideal, $\mathfrak{q}$ an $\mathfrak{m}$-primary ideal, M a finitely-generated A-module, (M_n) a stable $\mathfrak{q}$-filtration of M. Then*

(i) *M/M_n is of finite length for each $n \geqslant 0$;*

(ii) *for all suficiently large n this length is a polynomial $\chi_{\mathfrak{q}}^{M}(n)$ of degree $\leqslant s$, where s is the least number of generates of $\mathfrak{q}$;*

(iii) *the degree and leading coefficient of $\chi_{\mathfrak{q}}^{M}(n)$ depend only on M and $\mathfrak{q}$, not on the filtration chosen.*

If $M = A$, the polynomial $\chi_{\mathfrak{q}}^{M}(n)$ is called the characteristic polynomial of the $\mathfrak{m}$-primary ideal $\mathfrak{q}$, simply noted as $\chi_{\mathfrak{q}}(n)$.

Proposition 11.6. $\deg \chi_{\mathfrak{q}}(n) = \deg \chi_{\mathfrak{m}}(n)$.

DIMENSION THEORY OF NOETHERIAN LOCAL RINGS

Theorem 11.14 (Dimension theorem)**.** *For any Noetherian local ring A the following three integers are equal:*

(i) *the maximum length of chains of prime ideals in A (Krull dimension);*

(ii) *the degree of the characteristic polynomial $\chi_{\mathfrak{m}} = l\left(A/\mathfrak{m}^n\right)$;*

(iii) *the least number of generators of an $\mathfrak{m}$-primary ideal of A.*

Corollary 11.17 (Krull's principal ideal theorem)**.** *Let A be a Noetherian ring and let x be an element of A which is nether a zero-divisor nor a unit. Then every minimal prime ideal $\mathfrak{p}$ of (x) has height 1.*

REGULAR LOCAL RINGS

Theorem 11.22. *Let A be a Noetherian local ring of dimension d, $\mathfrak{m}$ its maximal ideal, $k = A/\mathfrak{m}$. Then the following are equivalent:*

(i) $G_{\mathfrak{m}}(A) \cong k[t_1, \cdots, t_d]$ *where the t_i are independent indeterminates;*

(ii) $\dim_k \left(\mathfrak{m}/\mathfrak{m}^2\right) = d$*;*

(iii) $\mathfrak{m}$ *can be generated by d elements.*

The ring satisfying any of the conditions is called regular local ring.

Proposition 11.23. *Let A be a ring, $\mathfrak{a}$ an ideal of A such that $\bigcap_n \mathfrak{a}^n = 0$. Suppose that $G_{\mathfrak{a}}(A)$ is an integral domain. Then A is an integral domain.*

Proposition 11.24. *Let A be a Noetherian local ring. Then A is regular iff $\hat{A}$ is regular.*

TRANSCENDENTAL DIMENSION

Assume that k is an algebraically closed field and let V be an irreducible affine variety over k. Let the coordinate ring of V be $A(V) = k[x_1, \cdots, x_n]/I(V)$. Then $k(V) = \operatorname{Frac} A(V)$ is a finitely generated extension of k and so has a finite transcendence degree over k, which is defined to be the dimension of V.

If P is a point with maximal ideal $\mathfrak{m}$ we call the Krull dimension $\dim A(V)_{\mathfrak{m}}$ the local dimension of V at P.

Theorem 11.25. *For any irreducible variety V over k the local dimension at any point is equal to its dimension.*

EXERCISES

1 Rings and Ideals

1.1. Consider $(1+x)\left(1-x+x^2-\cdots+(-x)^{n-1}\right)$.

1.2. (i) Passing to $(A/\mathfrak{p})[x]$. $A/\mathfrak{p}$ is an integral domain, so $\deg\left(\overline{f}\overline{g}\right)=\deg\left(\overline{f}\right)+\deg\left(\overline{g}\right)$. $\overline{f}\overline{g}=1$ means $a_1,\cdots,a_n\in\mathfrak{p}$. Use Proposition 1.8.
(ii) a, b are nilpotents $\Rightarrow a+b$ is nilpotent. Note that $f-a_0=xf_1$ is nilpotent.
(iii) Let $gf=0$ and g is of minimal degree. The leading coefficient $a_nb_m=0$, so a_ng has less degree than g and so $a_ng=0$. Same process to the second leading and following coefficients, we have $a_ig=0$ for all i.
(iv) (Gauss's lemma) Let $\mathfrak{a}$, $\mathfrak{b}$ and $\mathfrak{c}$ are generated by the coefficients of f, g and fg, respectively. We need to proof $\mathfrak{c}=(1)\Leftrightarrow\mathfrak{ab}=(1)$. $\mathfrak{c}\subseteq\mathfrak{ab}$ by the form of coefficients in fg. If $\mathfrak{c}\subsetneq\mathfrak{ab}=(1)$, let $\mathfrak{m}$ be the maximal ideal containing $\mathfrak{c}$, then $(A/\mathfrak{m})[x]$ is an integer domain. $\overline{fg}=0$ but $\overline{f},\overline{g}\neq 0$.

1.3. Induction on the indeterminates.

1.4. Pick the y in Proposition 1.9 to be the indeterminate, then use Exercise 1.2(i).

1.5. (i) $\Leftarrow$: $fg=\sum_{i=0}^{\infty}\left(\sum_{j=0}^{i}a_jb_{i-j}\right)x^i$, so we can find g such that $\sum_{j=0}^{i}a_jb_{i-j}=0$.
(ii) Similar to Exercise 1.2(ii). Construct a counterexample where a_n have increasing nilpotencies.
(iii) Pick the y in Proposition 1.9 to be 1. Use (i) on $1-f\Leftrightarrow 1-a_0$.
(iv),(v) $A\cong A[[x]]/(x)$. Use Proposition 1.1.

1.6. $\mathfrak{J}$ is an ideal, so there is an idempotent $e\in\mathfrak{J}$ but $e\notin\mathfrak{R}$. Pick the y in Proposition 1.9 to be 1.

1.7. Note that $x\left(1-x^{n-1}\right)=0\in\mathfrak{p}$.

1.8. Zorn's lemma but define the partial order reversely.

1.9. $\Rightarrow$: Apply Proposition 1.8 to $A/\mathfrak{a}$.
$\Leftarrow$: Use Proposition 1.11(ii).

1.10. Use Corollary 1.5 and Corollary 1.6(i)

1.11. (i) $x+x=(x+x)^2=x^2+2x+x^2=x+2x+x$.
(ii) Proof that every element in $A/\mathfrak{p}$ is 1.
(iii) $x+y-xy$.

1.12. Use Corollary 1.5.

1.13. 1 is a monic polynomial but it is neither irreducible nor reducible. A is an integral domain, so $\deg(fg)=\deg(f)+\deg(g)$. If $\mathfrak{a}=(1)$, apply the equation to $1=\sum_f y_f f(x_f)$.

1.14. If $xy\in\mathfrak{m}$, xy is zero-divisor, so are x and y. By maximality, $\mathfrak{m}\subseteq\mathfrak{m}+(x)=\mathfrak{m}$.

1.15. (i) Apply Proposition 1.8 to $A/\mathfrak{a}$ to proof the second equation.
(iv) Use $\sqrt{\mathfrak{a}\cap\mathfrak{b}}=\sqrt{\mathfrak{ab}}$.

1.16. Irreducible polynomials relate to prime ideals.

1.17. (i)-(iv) Use Exercise 1.15.
(v) Let $(X_{f_i})_{i\in I}$ be an open cover. By Exercise 1.15, $V\left(\{f_i\}_{i\in I}\right)=\varnothing$, so $\{f_i\}_{i\in I}$ generates the unit ideal. Let $1=\sum_{i\in J}g_if_i$ where J is a finite subset of I. If $\mathfrak{p}\notin\bigcup_{i\in J}X_{f_i}$, then $f_i\in\mathfrak{p}$ for all $i\in J$.

(vi) Similar to (v), where the unit ideal shall change to the radical $\sqrt{(f)}$.

1.18. (iv) If $V(\mathfrak{p}_x) \subseteq V(\mathfrak{p}_y)$, $\mathfrak{p}_x \in \overline{V(\mathfrak{p}_y)}$. If they don't have containing relation, $\mathfrak{p}_x \in \overline{V(\mathfrak{p}_y)}$ and $\mathfrak{p}_y \in \overline{V(\mathfrak{p}_x)}$.

1.19. Use the closed sets definition. Consider minimal prime ideals.

1.20. (i) Let $\overline{Y} = Z_1 \cup Z_2$, then $Z_i \cap Y = \varnothing$ for an $i \in \{1, 2\}$. Thus $Z_i = \overline{Z_i \cap Y} = \varnothing$.
(ii) Similar to (i). Construct the maximal element and proof that it is irreducible.
(iii) If Y Hausdorff but irreducible, every pair of open sets in Y intersect. Thus Y is singleton.
(iv) Use the closed set definition.

1.21. Simplify the sets: $\phi^{*-1}(E) = \{\mathfrak{q} \in Y : \phi^{-1}(\mathfrak{q}) \in E\}$, $\phi^*(V(E)) = \{\phi^{-1}(\mathfrak{q}) : E \subseteq \mathfrak{q}\}$.

1.22. Use quotient to find the form of prime ideals in A.

1.23. (i) $X_f = V(1-f)$.
(ii) $X_{f_1} \cup X_{f_2} = X_{f_1+f_2-f_1f_2}$.
(iii) A closed set in compact space is compact. By Exercise 1.17, it is a finite union of basic open set. Then apply (ii).
(iv) By Exercise 1.11 every prime ideal is maximal. Pick $f \in \mathfrak{p}_x \backslash \mathfrak{p}_y$, then X_f and $V(f)$ separate two points.

1.24. $a = a \wedge 1 = a \wedge (a \vee a') = (a \wedge a) \vee (a \wedge a') = (a \wedge a) \vee 0 = a \wedge a$.

1.25. Use Exercise 1.23(iv).

1.26. $\mathfrak{m}_x \in \mu(U_f) \Leftrightarrow f(x) \neq 0 \Leftrightarrow f \notin \mathfrak{m}_x \Leftrightarrow \mathfrak{m}_x \in \tilde{U}_f$.

1.27. If $x \neq y$, there exists a polynomial $f \in P(X)$ such that $f(x) = 0$ but $f(y) \neq 0$, so $\mathfrak{m}_x \neq \mathfrak{m}_y$, and so μ injective. By Nullstellensatz, every maximal ideal in $P(X)$ is of the form $\mathfrak{m} = (t_1 - a_1, \cdots, t_n - a_n)$, with $(a_1, \cdots, a_n) \in X$, so μ surjective. By Exercise 1.26, μ is a homeomorphism.

1.28. Project to coordinate functions.

2 Modules

2.1. Bézout's identity.

2.2. Tensor the exact sequence $0 \to \mathfrak{a} \to A \to A/\mathfrak{a} \to 0$. Calculate the image of $\mathfrak{a} \otimes M \to A \otimes M$.

2.3. Consider the residue field k of A and pass $M \otimes_A N$ to vector space $M_k \otimes_k N_k$. Use Exercise 2.2 and Nakayama's lemma.

2.4. There exist canonical projections.

2.5. The definition of flatness only considers the module structure. Use Exercise 2.4.

2.6. Construct two homomorphisms.

2.7. Proof that $(A/\mathfrak{p})[x]$ and $A[x]/\mathfrak{p}[x]$ are isomorphic.

2.8. (ii) Use Proposition 2.15.

2.9. Pick $y_1, \cdots, y_m \in M$ whose images are the generators of M''. Use diagram chasing for an element of M.

2.10. $u(m) + \mathfrak{a}N \in N$, then use surjectiveness. Apply Nakayama's lemma on $(u(M) + \mathfrak{a}N)/N$.

2.11. (i),(ii) Consider a residue field and pass to vector spaces. Use the right exactness of tensor product.

(iii) If $m > n$, let $\psi : A^m \xrightarrow{\phi} A^n \hookrightarrow A^m$. By Proposition 2.4, ψ satisfies $\psi^n + a_{n-1}\psi^{n-1} + \cdots + a_0 = 0$. Pick such equation to have the minimum degree, then ψ injective implies $a_0 \neq 0$. Apply the equation to element $(0, \cdots, 0, 1)$ to obtain contradiction.

2.12. Choose $u_i \in M$ such that $\phi(u_i)$ are a basis of A^n, then proof that $0 \to \ker(\phi) \to M \xrightarrow{\phi} A^n \to 0$ splits.

2.13. $p \circ g = \mathrm{id}_N$ injective $\Rightarrow$ g injective, so $0 \to N \xrightarrow{g} N_B \to N_B/g(N) \to 0$ splits.

2.14. This question asks to proof $\mu_i = \mu_j \circ \mu_{ij}$.

2.15. (i) Use Exercise 2.14 to increase indices.
(ii) $x_i = \sum_k (y_k - \mu_{kl}(y_k))$ is an equation in $\bigoplus M_i$, so μ_{ij} cannot be applied directly. Use canonical projection $\pi_i : \bigoplus M_i \to M_i$ on the equation first.

2.16 (Universal property of direct limit)**.** Use Exercise 2.15(i) for both existence and uniqueness.

2.17. Similar to Exercise 2.15(i), move components into a same index.

2.18. Use Exercise 2.16.

2.19 ($\varinjlim$ is an exact functor)**.** Diagram chasing. Use Exercise 2.15.

2.20. Draw the diagram. Define two homomorphisms by the universal properties of tensor product and direct limit. Verify their compositions are identity maps.

2.21. Use Exercise 2.15(ii) on $\alpha_i(1)$.

2.22. Use Exercise 2.15. Pick a nilpotent or assume $xy = 0$.

2.23. If $J \subseteq J'$, $B_J \cong B_J \otimes (A)_{i \in J' \setminus J}$. Let Γ be the set of all finite subsets of Λ, then $(B_J)_{J \in \Gamma}$ is the case of Exercise 2.21. We can also use Exercise 2.17 to see its form.

2.24. Directly from the definition of Tor functor.

2.25. Use Snake lemma.

2.26. Let M_n be generated by $\{x_1, \cdots, x_n\}$. We have an exact sequence

$$\mathrm{Tor}_1(M_{n-1}, N) \to \mathrm{Tor}_1(M_n, N) \to \mathrm{Tor}_1(M_n/M_{n-1}, N).$$

By Proposition 2.3, $M_n/M_{n-1} \cong A/\mathfrak{a}$.

Let $f : \mathfrak{a} \otimes N \to A \otimes N$, $f\left(\sum a_i \otimes y_i\right) = 0$. Let $\mathfrak{a}'$ be generated by a_i. By assumption, $f|_{\mathfrak{a}' \otimes N}$ is injective, so $\sum a_i \otimes y_i = 0$ and thus $\mathrm{Tor}_1(A/\mathfrak{a}, N) = 0$. Since $M_0 = A$ and $\mathrm{Tor}_1(A, N) = 0$, the exact sequence implies $\mathrm{Tor}_1(M_n, N) = 0$ for all n. Then use Proposition 2.19(iv).

Another approach is to use the fact that any module is the direct limit of some finitely generated modules, and direct limit is an exact functor.

2.27. (i) $\Rightarrow$ (ii) $A/(x)$ is a flat A-module, then $\phi : A/(x) \otimes (x) \to A/(x)$ injective. $\phi(1 \otimes x) = x = 0$, so $A/(x) \otimes (x) = 0$. Then use Exercise 2.2.
(ii) $\Rightarrow$ (iii) Let $x = ax^2$, then ax is an idempotent. $(x) \ni bx = bax^2 = (bx)(ax) \in (ax)$, so $(ax) = (x)$. Similar to Exercise 1.11(iii), every finitely generated ideal is generated by an idempotent e, so $A = (e) \oplus (1 - e)$.
(iii) $\Rightarrow$ (i) $N \otimes_A A = (N \otimes_A \mathfrak{a}) \oplus (N \otimes_A \mathfrak{a}')$, so $(N \otimes_A \mathfrak{a}) \to N \otimes_A A$ injective. Use Exercise 2.26.

2.28. Use Exercise 2.27(ii).

3 Rings and Modules of Fractions

3.1. The only if part requires the finitely generated property.

3.2. Use Proposition 1.9. Construct the inverse by elementary algebra.
If $M = \mathfrak{a}M$, $S^{-1}M = S^{-1}\mathfrak{a}$, $S^{-1}M = \left(S^{-1}\mathfrak{a}\right)\left(S^{-1}M\right)$. Then use Nakayama's lemma and Exercise 3.1.

3.3. Construct the isomorphism by $(a/s)/(t/1) \mapsto a/(st)$.

3.4. Construct the isomorphism.

3.5. If $a \in A$ is a nilpotent, let $\mathfrak{p} \supseteq \operatorname{Ann}(a)$. $a/1 \in A_\mathfrak{p}$ is zero, so $sa = 0$ for some $s \in A - \mathfrak{p}$.
Counter example is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

3.6. Use the fact that for any multiplicatively closed set S, if $A - S$ is an ideal, it's automatically prime.

3.7. (i),(ii) More specifically, $A - \overline{S} = \bigcup_{\mathfrak{p}\cap S=\varnothing} \mathfrak{p}$.
(iii) $\mathfrak{p} \cap (1 + \mathfrak{a}) = \varnothing \Leftrightarrow \mathfrak{a} + \mathfrak{p} \neq (1) \Leftrightarrow \mathfrak{a} + \mathfrak{p} \subseteq \mathfrak{m}$ for some maximal ideal $\mathfrak{m}$. Note that the union of a chain of prime ideals is only a maximum ideal.

3.8. (v) $\Rightarrow$ (ii) If $t/1$ is not a unit in $S^{-1}A$, it is contained in a prime ideal $\mathfrak{q} \subseteq S^{-1}A$, so $t \in T \cap \mathfrak{q}^c$.

3.9. Let S be maximal in Exercise 3.6. If $S_0 \nsubseteq S$, $S \subsetneq SS_0$. Then use Exercise 3.6.
(i) Assume that S_1 is larger, then pick a zero-divisor $x \in S_1$.
(ii) If a/s is not a zero-divisor, a is not a zero-divisor, so $a \in S_0$.
(iii) Note that $s \in S_0$ is a unit.

3.10. (i) Exercise 2.27 says for any element $x \in A$, $x = ax^2$ for some $a \in A$.
(ii) $\Rightarrow$: Use Exercise 2.28. $\Leftarrow$: Consider an $A_\mathfrak{m}$-module. Use Proposition 3.10.

3.11. (i) $\Rightarrow$ (ii) Use Exercise 2.27 on $A/\mathfrak{p}$.
(ii) $\Rightarrow$ (i) Proof that $(A/\mathfrak{R})_{\overline{\mathfrak{p}}}$ has no nilpotent, then use Exercise 1.10 and Exercise 3.10.
(ii) $\Rightarrow$ (iv) Let $f \in \mathfrak{p}_1$ but $f \notin \mathfrak{p}_2$, then $f/1 \in A_{\mathfrak{p}_1}$ is nilpotent, so $sf^n = 0$. We have $\mathfrak{p}_1 \in X_s$ and $\mathfrak{p}_2 \in X_{f^n}$. By Exercise 1.17, $X_s \cap X_{f^n} = X_{sf^n} = \varnothing$.

3.12. (i)-(iii) Use the torsion element definition.
(iv) Use Proposition 3.5.

3.13. Proof they contain in each other. Use Proposition 3.8 for the equivalent conditions.

3.14. Pass to $M/\mathfrak{a}M$. Use Proposition 3.8.

3.15. Let $\{x_i\}_{i=1}^n$ be generators and $\{e_i\}_{i=1}^n$ be the canonical basis. Let $\phi(e_i) = x_i$, so ϕ is surjective. Consider a residue field $k = A/\mathfrak{m}$, we have $\operatorname{Tor}_1(k, F) \to k \otimes \ker\phi \to k \otimes F \xrightarrow{1\otimes\phi} k \otimes F \to 0$. $\operatorname{Tor}_1(k, F) = 0$ by balancing Tor theorem; $1 \otimes \phi$ bijective so $k \otimes \ker\phi = 0$. By Exercise 2.3, $\ker\phi = 0$.
If $\{x_i\}_{i=1}^m$ generate F where $m < n$, pick some element $\{x_i\}_{i=m+1}^n$ so we also have $\phi(e_i) = x_i$. By injectiveness of ϕ, $\{e_i\}_{i=1}^n$ are linearly dependent.

3.16. (i) $\Rightarrow$ (ii) $f^*(p^e) = p$.
(ii) $\Rightarrow$ (iii) Use Proposition 1.17.
(iii) $\Rightarrow$ (iv) Every module contains a cyclic submodule, which is isomorphic to $A/\mathfrak{a}$. Then use $B \otimes_A A/\mathfrak{a} \cong B/\mathfrak{a}^e$ and $\mathfrak{a}^e \subseteq \mathfrak{m}^e$.
(iv) $\Rightarrow$ (v) Let the mapping be ϕ. Since B flat, $0 \to (\ker\phi)_B \to M_B \to (M_B)_B$ exact. By Exercise 2.13, $M_B \to (M_B)_B$ injective, so $(\ker\phi)_B = 0$; by assumption, $\ker\phi = 0$.
(v) $\Rightarrow$ (i) Take $M = A/\mathfrak{a}$. Then $M_B = B/\mathfrak{a}^e$.

3.17. If $M \to N$ injective, we have a diagram

$$\begin{tikzcd} B \otimes_A M \arrow[r, "\phi"] \arrow[d, "g\otimes \mathrm{id}_M"] & B \otimes_A N \arrow[d, "g\otimes \mathrm{id}_N"] \\ C \otimes_A M \arrow[r, "\psi"] & C \otimes_A N \end{tikzcd}$$

$g \circ f$ flat, so ψ injective. g faithfully flat, by Exercise 3.16(v), $B \otimes_A M \to C \otimes_B (B \otimes_A M)$ injective. By Proposition 2.15, $C \otimes_B (B \otimes_A M) = C \otimes_A M$, so $g \otimes \mathrm{id}_M$ injective.

3.18. Use Exercise 3.3 to proof $B_\mathfrak{q} \cong (B_\mathfrak{p})_{\mathfrak{q}B_\mathfrak{p}}$. By Proposition 3.3 and Proposition 3.5, $B_\mathfrak{q}$ is flat over $B_\mathfrak{p}$. Hence $B_\mathfrak{q}$ is flat over $A_\mathfrak{p}$ and satisfies Exercise 3.16(iii).

3.19. (v) $M \cong \bigoplus_{i=1}^n A/\mathfrak{a}_i$, so $\mathrm{Supp}(M) = \bigcup_{i=1}^n \mathrm{Supp}\,(A/\mathfrak{a}_i) = \bigcup_{i=1}^n V(\mathfrak{a}_i) = V\left(\bigcap_{i=1}^n \mathfrak{a}_i\right)$. Proof that $\bigcap_{i=1}^n \mathfrak{a}_i = \mathrm{Ann}(M)$.
(vi) Use Exercise 2.3.
(vii) Use Exercise 2.2.
(viii) $\mathfrak{q} \in f^{*-1}(\mathrm{Supp}(M)) \Leftrightarrow M_{\mathfrak{q}^c} \neq 0 \Leftrightarrow M_\mathfrak{p} \neq 0$ and $B_\mathfrak{q} \neq 0 \Leftrightarrow \mathfrak{q} \in \mathrm{Supp}(M) \cap \mathrm{Supp}(B)$.

3.20. Counter example is $A[x]/\left(x^2\right)$.

3.21. (i),(ii) Use Proposition 3.11(iv).
(iii) Use Exercise 1.21(iv).
(iv) $A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p} = \mathrm{Frac}(A/\mathfrak{p})$. Use Proposition 2.15 on $A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p} \otimes_{A/\mathfrak{p}} A/\mathfrak{p} \otimes_A B$ to proof the equation.

3.22. Let $\mathfrak{p} \in X_f$, so $f \notin \mathfrak{p}$. Then $\mathfrak{q} \in \mathrm{Spec}\,(A_\mathfrak{p}) \Leftrightarrow \mathfrak{q} \subseteq \mathfrak{p} \Leftrightarrow f \notin \mathfrak{q} \Leftrightarrow \mathfrak{q} \in X_f$.

3.23. (i) Use Exercise 1.17(iv).
(v) Define two homomorphisms by the universal properties of direct limit and local ring.

3.24.

3.25.

3.26.

3.27.

3.28.

3.29.

3.30.

4 Primary Decomposition

4.1. Use Exercise 1.21(iv), so $\mathrm{Spec}(A/\mathfrak{a}) = V(\mathfrak{a})$.

4.2. All primary components are prime.

4.3. For any element $x \in A$, $x(1 - ax) = 0 \in \mathfrak{q}$.

4.4. $\mathfrak{m} = \sqrt{\mathfrak{q}} \supsetneq \mathfrak{q} \supsetneq \mathfrak{m}^2$.

4.5. $\mathfrak{m}^2$ is $\mathfrak{m}$-primary.

4.6.

4.7. (ii),(iv),(v) $(\mathfrak{a}[x])^c = \mathfrak{a}$.
(iii) Let $\overline{f}, \overline{g} \in (A/\mathfrak{q})\,[x]$. If $\overline{f}\overline{g} = 0$ but $\overline{g} \neq 0$, by Exercise 1.2(iii), $\overline{a}\overline{f} = 0$, so $aa_i \in \mathfrak{q}$. Hence $a_i \in \mathfrak{p}$ and $a_i^{n_i} \in \mathfrak{q}$, so $\overline{a_i}^{n_i} = 0$ and use Exercise 1.2(ii).

4.8. $k[x_1,\cdots,x_n]/\mathfrak{p}_\mathfrak{i} = k[x_{i+1},\cdots,x_n]$ integer domain, so $\mathfrak{p}_i$ prime. Note that $\mathfrak{p}_i^n$ is the set of all homogeneous polynomials in $k[x_1,\cdots,x_i]$ of degree $\geqslant n$.

4.9. (i) $\Leftarrow$: x is contained in all prime ideals containing $(0:a)$, so $x \in \sqrt{(0:a)}$.
(ii) If $\mathfrak{a} \subseteq \mathfrak{p}$ but $\mathfrak{a}^e \subseteq S^{-1}\mathfrak{p}' \subsetneq S^{-1}\mathfrak{p}$, then $\mathfrak{a} \subseteq \mathfrak{a}^{ec} \subseteq \mathfrak{p}' \subsetneq \mathfrak{p}$. If $\mathfrak{b} \subseteq S^{-1}\mathfrak{p}$ but $\mathfrak{b}^c \subseteq \mathfrak{p}' \subsetneq \mathfrak{p}$, use $\mathfrak{b}^{ce} = \mathfrak{b}$ derived from Proposition 3.11(i). $S^{-1}\mathfrak{p}$ is a prime ideal if S doesn't meet $\mathfrak{p}$.
(iii) $(0:a) = \bigcap(\mathfrak{q}_i : a) \subseteq \mathfrak{p}$. By Proposition 1.11(ii), $(\mathfrak{q}_i : a) \subseteq \mathfrak{p}$ for some i. Then use Proposition 4.4 and the minimality of $\mathfrak{p}$.

4.10. (iv) Let $x \neq 0$, $(0:x) \subseteq \mathfrak{p}$ for some $\mathfrak{p}$. If $x \in S_\mathfrak{p}(0)$, then $sx = 0$, so $s \in (0:x)$.

4.11. Note that $x \in S(\mathfrak{a}) \Leftrightarrow$ there exists an $s \in S$ such that $sx \in \mathfrak{a}$. Show that $S_\mathfrak{p}(0)$ is contained in all $\mathfrak{p}$-primary ideals.
If $0 = \bigcap \mathfrak{q}_i$ is irredundant, $\sqrt{(0)}$ is the intersection of minimal prime ideals.

4.12. Use Proposition 3.11(ii). Use Proposition 4.9 to show there are at most 2^n different $S(\mathfrak{a})$.

4.13. (i) Use Exercise 4.12.
(ii) Use Proposition 4.8 and Proposition 4.9.
(iii) Proof that $S_\mathfrak{p}\left(\mathfrak{p}^{(m)}\mathfrak{p}^{(n)}\right) = \mathfrak{p}^{(m+n)}$.
(iv) $\Leftarrow$: Show that $\mathfrak{p}^{(n)}$ is the smallest $\mathfrak{p}$-primary ideal containing $\mathfrak{p}^n$.

4.14. Let $x \notin \mathfrak{q}$. If $ab \in (\mathfrak{q}:x)$ but $a \notin (\mathfrak{q}:x)$, we have $xab \in \mathfrak{q}$ but $xa \notin \mathfrak{q}$. Thus, $b \in (\mathfrak{q}:xa) = (\mathfrak{q}:x)$ by maximality, and therefore $(\mathfrak{q}:x)$ is prime.

4.15. Use Proposition 4.9.

4.16. Use Proposition 3.11(iv) and Proposition 4.9.

4.17.

4.18. (i) $\Rightarrow$ (L1) By Proposition 4.9, $S_\mathfrak{p}(\mathfrak{a}) = \bigcap \mathfrak{q}_i$. Pick $x \in S_\mathfrak{p}$, then $\mathfrak{q}_i = (\mathfrak{q}_i : x)$ by Proposition 4.4.
(i) $\Rightarrow$ (L2) Use Exercise 4.12.

4.19. Similar to the proof of Exercise 4.11.
The goal is to proof that there exists $\mathfrak{a} = \bigcap_{i=1}^n \mathfrak{q}_i$ irredundant, so we need to proof $\mathfrak{q}_i \not\supseteq \bigcap_{j\neq i}\mathfrak{q}_j$. Using the fact that $\mathfrak{u}\mathfrak{v} \subseteq \mathfrak{q}$ but $\mathfrak{u} \not\subseteq \mathfrak{q} \Rightarrow \mathfrak{v} \subseteq \mathfrak{p}$, we can assume $\mathfrak{b} = \bigcap_{i=1}^{n-1}\mathfrak{q}_i$ irredundant and $\mathfrak{p}_n$ maximal among $\{\mathfrak{p}_i\}$, so we only need to find $\mathfrak{q}_n \not\supseteq \mathfrak{b}$. If such $\mathfrak{q}_n$ doesn't exist, $\mathfrak{b} \subseteq S_{\mathfrak{p}_n}(0)$. Pick a minimal prime ideal $\mathfrak{p} \subseteq \mathfrak{p}_n$ and use Exercise 4.10 and Proposition 1.11(ii).

4.20.

4.21.

4.22.

4.23.

5 Integral Dependence and Valuations

5.1. $f^*(V(I)) = \left\{f^{-1}(\mathfrak{q}) : \mathfrak{q} \supseteq I\right\}$. Use Theorem 5.10 to proof $f^*(V(I)) \supseteq V\left(f^{-1}(I)\right)$.

5.2. $\mathfrak{p} = \ker f$ is prime. By Theorem 5.10, there is a prime ideal $\mathfrak{q} \subseteq B$. $\mathrm{Frac}(B/\mathfrak{q})$ is an algebraic extension of $\mathrm{Frac}(A/\mathfrak{p})$, so $\mathrm{Frac}(A/\mathfrak{p}) \to \Omega$ can be extended to $\mathrm{Frac}(B/\mathfrak{q}) \to \Omega$.

5.3. Use the operator $- \otimes_A c^n$.

5.4. If $\left(\frac{1}{x+1}\right)^n + \sum \frac{f_i\left(x^2-1\right)}{g_i(x^2-1)(x+1)^i} = 0$, multiply both side by $(x+1)^n$ and take $x = -1$ to get contradiction.

5.5. (ii) Use Theorem 5.10 to proof that the contraction of maximal ideals in B are maximal in A.

5.6. Let $b_i \in B_i$, $f_i \in A[x_i]$ monic, $f_i(b_i) = 0$. Let $f = (f_1, \cdots, f_n)$, then $f(b_1, \cdots, b_n) = 0$.

5.7. Let $x \in B$ and x integral over A. Let $x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$ is the equation of the least degree. Analyze $x\left(x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1\right) = -a_0$.

5.8. Proof that there is a ring $D \supseteq B$ such that f and g factor into linear factors in D. Use Kronecker's construction.

5.9. Let $f^n + g_{n-1}f^{n-1} + \cdots + g_0 = 0$. Let $f_1 = f - x^r$ with r larger than all $\deg g_i$, then $f_1\left(f_1^{n-1} + h_{n-1}f_1^{n-2} + \cdots + h_1\right) = -h_0$ with h_0 monic. Then use Exercise 5.8.

5.10. (a) $\Rightarrow$ (b) Let $f^*(V(\mathfrak{q}_1)) = V(I)$, so $I \subseteq \mathfrak{p}_1 \subseteq \mathfrak{p}_2$, and we have $\mathfrak{p}_2 \in V(I) = f^*(V(\mathfrak{q}_1))$.
(a') $\Rightarrow$ (c')

5.11. Use Exercise 3.18 and Exercise 5.10.

5.12. Note that G has identity element so x is a root of the polynomial $\prod_{\sigma\in G}(t - \sigma(x))$. The coefficients are symmetric polynomials of σ_i. G acting on $S^{-1}A$ is defined to be $\sigma(x/s) = \sigma(x)/\sigma(s)$.

5.13. Let $\mathfrak{q} \in P$. Then $\sigma(\mathfrak{q}) \cap A^G = \mathfrak{q} \cap A^G = \mathfrak{p}$, so $\sigma(\mathfrak{q}) \in P$. Let $\mathfrak{q}_1$, $\mathfrak{q}_2 \in P$ and $x \in \mathfrak{q}_1$. Then $\prod_{\sigma\in G}\sigma(x) \in \mathfrak{q}_1 \cap A^G = \mathfrak{p} \subseteq \mathfrak{q}_2$, so $x \in \sigma^{-1}(\mathfrak{q}_2)$, and so $\mathfrak{q}_1 \subseteq \bigcup_{\sigma\in G}\sigma(\mathfrak{q}_2)$. Use Proposition 1.11(i) and Corollary 5.9.

5.14. Note that $\sigma \in G$ fixes A.

5.15.

5.16.

5.17. Consider $k[t_1, \cdots, t_n]/I(X)$. Use Exercise 5.16.

5.18 (Zariski's lemma)**.** Let n be the number of generators. If $n = 1$, x_1 is invertible, so $x_1^{-1} = \sum_{i=1}^n c_i x_1^i$.
If $n > 1$, let $A = k[x_1]$ and $K_A = \operatorname{Frac}(A)$. Then $K_A[x_2, \cdots, x_n]$ is a finite algebraic extension of K_A, so there are monic polynomials $f_i \in K_A[y]$ such that $f_i(x_i) = 0$ for $2 \leqslant i \leqslant n$. Pick $a \in A$ such that $af_2 \cdots f_n \in A[y]$, then $x_2, \cdots, x_n$ are integral over A_a. Therefore, $K_A \subseteq B$ are integral over A_a.
If x_1 transcendental over k, A is a UFD. Since every UFD is integrally closed, A is integrally closed, so A_a is integrally closed by Proposition 5.12. K_A integral over A_a, so $A_a = K_A$, but such a must be a unit in A.

5.19. By Exercise 5.18, $k[t_1, \cdots, t_n]/\mathfrak{m}$ is a finite algebraic extension of k, so $k[t_1, \cdots, t_n]/\mathfrak{m} \cong k$. Let $t_i \mapsto a_i$, then $t_i - a_i \in \mathfrak{m}$. Proof that $(t_1 - a_1, \cdots, t_n - a_n)$ is maximal.

5.20. Hint.

5.21. We have

$$\begin{tikzcd}
A \arrow[r, "\phi_A"] \arrow[drrr, "f"'] & A_s \arrow[r] \arrow[drr, "f_s"'] & A_s[y_1, \cdots, y_n] \arrow[r] \arrow[dr, "f_s'"'] & B_s \arrow[d, "g"] & B \arrow[l, "\phi_B"'] \\
& & & \Omega &
\end{tikzcd}$$

f_s by universal property of localization. f_s' by defining $y_i \mapsto 0$. By Exercise 5.20, B_s integral over $A_s[y_1, \cdots, y_n]$, so g is obtained by Exercise 5.2.

5.22. Hint.

5.23. (i) $\Rightarrow$ (ii) Note that $f : A \to A/\ker f$. Use Proposition 1.1.
(ii) $\Rightarrow$ (iii) Consider $A/\mathfrak{p}$.

(iii) $\Rightarrow$ (i) If $\mathfrak{p}$ is not an intersection of maximal ideals, the Jacobson radical $\mathfrak{J}$ of $A/\mathfrak{p}$ is not zero. Pick $f \in \mathfrak{J}$, then $(A/\mathfrak{p})_f \neq 0$. The contraction of a maximal ideal of $(A/\mathfrak{p})_f$ is $\mathfrak{q} \in A$ which is not maximal and doesn't contain f, but all prime ideals strictly containing $\mathfrak{q}$ contains f.

5.24. (i) Use Exercise 5.23(iii) and going-up theorem.
(ii) Consider $B/\mathfrak{q}$ and $A/\mathfrak{q}^c$. Then use Exercise 5.22, Exercise 5.23(i) and Proposition 1.1.

5.25. Hint.

5.26. (2) $\Rightarrow$ (3) $U_1 \cap X_0 = U_2 \cap X_0 \Rightarrow U_1^c \cap X_0 = U_2^c \cap X_0$, then take closure.
(i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) Consider $V(\mathfrak{a}) \cap X_f \cap \operatorname{Max}(A)$. If all maximal ideals in $V(\mathfrak{a})$ contains f, $V(\mathfrak{a}) \cap X_f = \varnothing$ by Exercise 5.23(iii).
(iii) $\Rightarrow$ (i) If not Jacobson, there is a prime ideal $\mathfrak{p}$ such that $\mathfrak{p} \subsetneq \bigcap_{\mathfrak{q} \supseteq \mathfrak{p}} \mathfrak{q}$. Let $f \in \bigcap_{\mathfrak{q} \supseteq \mathfrak{p}} \mathfrak{q} \Big\backslash \mathfrak{p}$, then $X_f \cap V(\mathfrak{p}) = \{\mathfrak{p}\}$.

5.27. $\Rightarrow$: Use Theorem 5.21 to proof maximal elements are valuation rings.
$\Leftarrow$: If $(A, \mathfrak{m}) \subsetneq (A', \mathfrak{m}')$, pick $x \in A' \backslash A$. Then $x^{-1} \in \mathfrak{m}$ because it is not a unit in A.

5.28. (1) $\Rightarrow$ (2) Pick $a \in \mathfrak{a} \backslash \mathfrak{b}$, and any $b \in \mathfrak{b}$. Consider a/b and b/a.
(2) $\Rightarrow$ (1) Consider $(a) \subseteq (b)$ or $(b) \subseteq (a)$, we have $a = xb$ or $b = ya$, so $y = x^{-1}$.

5.29. B is a local ring of A means $B = A_{\mathfrak{p}}$ for some prime ideal $\mathfrak{p}$. Let $\mathfrak{m}$ be the maximal ideal of B and let $\mathfrak{p} = \mathfrak{m} \cap A$. If $s \in A$ but $s \notin \mathfrak{p}$, then $s \in B$ and $s \notin \mathfrak{m}$, so s is a unit in B. Let $A_{\mathfrak{p}} \to B$ to be $x/s \mapsto s^{-1}x$, which is injective; $\mathfrak{p}A_{\mathfrak{p}} \subseteq B$, so B dominates $A_{\mathfrak{p}}$. By Exercise 5.28, $A_{\mathfrak{p}}$ is a valuation ring, then use Exercise 5.27.

5.30. $xy^{-1} \in A$ or $x^{-1}y \in A$, so $\geqslant$ is a total order. Also $(xz)(yz)^{-1} = xy^{-1}$.
v can be a representation map. If $v(x) \geqslant v(y)$, $(x+y)y^{-1} = xy^{-1} + 1 \in A$.

5.31. Let the set be A. Verify A is closed under addition and multiplication. By taking $x = y = 1$, $v(1) = 0$, so $v(1) = v(x) + v\left(x^{-1}\right)$, and so A is a valuation ring.

5.32.

5.33.

5.34.

5.35.

6 Chain Conditions

6.1. (i) Let $x \in \ker(u)$. Since u^n surjective, there is a $y \in M$ such that $u^n(y) = x$. But $u^{n+1}(y) = u(x) = 0$, means that $y \in \ker\left(u^{n+1}\right) = \ker\left(u^n\right)$.
(ii) Let $x \in M$. Then $u^n(x) \in \operatorname{im}\left(u^n\right) = \operatorname{im}\left(u^{n+1}\right)$, so there is a $y \in M$ such that $u^{n+1}(y) = u^n(x)$. But u^n injective, means that $u(y) = x$.

6.2. If M is not Noetherian, there is a submodule N which is not finitely generated. Construct a chain of finitely generated submodules by the generators of N.

6.3. Consider $0 \to N_2/\left(N_1 \cap N_2\right) \to M/\left(N_1 \cap N_2\right) \to M/N_2 \to 0$. Use Proposition 2.1.

6.4. Let $m_1, \cdots, m_n$ be generators of M, then $\mathfrak{a} = \bigcap \operatorname{Ann}\left(m_i\right)$. $A/\operatorname{Ann}\left(m_i\right) \cong Am_i \subseteq M$, then use Exercise 6.3.

6.5. Use the open set definition.

6.6. (i) $\Rightarrow$ (iii) Use Exercise 6.5.
(ii) $\Rightarrow$ (i) Consider a chain of ascending open sets in X.

6.7. Let Σ be the collection of closed subsets of X which are not finite union of irreducible closed subspaces. Then Σ has minimal element X_0. Proof that X_0 cannot be either irreducible or reducible.

6.8. The converse isn't true. The counter example is $k[x_1, \cdots]/\left(x_1^2, \cdots\right)$.

6.9. The irreducible components of $\operatorname{Spec}(A)$ is $V(\mathfrak{p})$ where $\mathfrak{p}$ minimal (Exercise 1.20(iv)).

6.10. Use Exercise 3.19(v).

6.11. By Exercise 6.7, $V(\mathfrak{b})$ is a finite union of $V(\mathfrak{q}_i)$, where $\mathfrak{q}_i$ minimal and containing $\mathfrak{b}$. Then use Exercise 5.10(c).

6.12. The converse isn't true. The counter example is an infinite product $\prod \mathbb{Z}/2\mathbb{Z}$.

7 Noetherian Rings

7.1.

7.2.

7.3. (i) $\Rightarrow$ (ii): Use Proposition 4.8 and Proposition 4.4. (ii) $\Rightarrow$ (iii): Let $S = \left\{1, x, x^2, \cdots\right\}$. Proof that $\bigcup(\mathfrak{a} : x^n) = S(\mathfrak{a})$. (iii) $\Rightarrow$ (i): Similar to the proof of Proposition 7.12.

7.4.

7.5. Use Exercise 5.12 and Proposition 7.8.

7.6. Hint.

7.7. Use a.c.c..

7.8. $A \cong A[x]/(x)$.

7.9. Hint.

7.10. $M[x] = M \otimes A[x]$, so it is a homomorphic image of $M \times A[x]$.

7.11. Counter example is an infinite product $\prod \mathbb{Z}/2\mathbb{Z}$.

7.12. By Exercise 3.16(i), $\mathfrak{a} \mapsto \mathfrak{a}^e$ is injective. Then use a.c.c..

7.13. Transfer the Noetherian property $B \to B_\mathfrak{p} \to B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p}$.

7.14 (Hilbert's Nullstellensatz)**.** The hint uses a method other than Rabinowitsch trick.

7.15. Hint.

7.16. Use Exercise 7.15.

7.17.

7.18.

7.19.

7.20. (i) $\Rightarrow$: Rigorous proof requires using induction on $\mathcal{F}_i$, where $\mathcal{F}_i$ is formed by applying complement i times on the topology $\tau = \mathcal{F}_0$, and satisfies $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}$. (ii) $\Rightarrow$: Use Exercise 1.19.

7.21. $\Rightarrow$: Use Exercise 7.20. $\Leftarrow$: Use hint. If $\overline{E \cap X_0} \neq X_0$, $E \cap \overline{E \cap X_0} \in \mathcal{F}$ by minimality, but $E \cap \overline{E \cap X_0} = E \cap X_0$. If $U_0 \subseteq E \cap X_0$, let $U_0 = U \cap X_0$ where U is open in X, then $E \cap X_0 = U_0 \cup (E \cap U^c \cap X_0)$.

7.22. If $U_0 \subseteq E \cap X_0$, use the notations similar to Exercise 7.21. $U^c \cap X_0$ is closed, so by minimality $E \cap U^c \cap X_0$ is open.

7.23. Hint.

7.24. Hint.

7.25. Use Exercises 5.11 and 7.24.

7.26.

7.27.

8 Artinian Rings

8.1. Use Proposition 7.16 on local ring $A_{\mathfrak{p}_i}$. Then apply the contraction. If $\mathfrak{q}_i$ is isolated, then $\mathfrak{p}_i$ is minimal.

8.2. (i) $\Rightarrow$ (ii) Use Proposition 8.1.
(iii) $\Rightarrow$ (i) Use Proposition 8.5.

8.3.

9 Discrete Valuation Rings and Dedekind Domain

10 Completion

10.1. $B = \bigoplus_{n\geqslant 0} \mathbb{Z}/p^n\mathbb{Z}$, so $A_k = \alpha^{-1}\left(p^k B\right) = \bigoplus_{n>k} \mathbb{Z}/p\mathbb{Z}$. The sequence $0 \to A \to B \xrightarrow{p\cdot} B \to 0$ is exact but $\hat{A} \to \hat{B} \xrightarrow{\hat{p}\cdot} \hat{B} \to 0$ is not, due to the kernel of $\hat{p}\cdot$ is $\prod_{k\geqslant 0} \mathbb{Z}/p\mathbb{Z}$.

10.2. The explicit form of the given exact sequence is

$$0 \to \bigoplus_{k>n} \mathbb{Z}/p\mathbb{Z} \to \bigoplus_{n\geqslant 0} \mathbb{Z}/p\mathbb{Z} \to \bigoplus_{n=0}^{k} \mathbb{Z}/p\mathbb{Z} \to 0.$$

Then use Exercise 10.1. Analyze definitions in the proof of Proposition 10.2 to claim that $\operatorname{coker} d^B = \varprojlim^1 A = 0$.

10.3.

11 Dimension Theory